

CONTACT

- [email protected]
- linkedin.com/in/paddyiyer
- paddyspeaks.com
- github.com/paddyiyer
- San Francisco Bay Area

TECHNICAL SKILLS

Spark

Hive

Hadoop

Databricks

Delta Lake

Snowflake

Redshift

dbt

Airflow

Datasmarm

AWS

Azure

GCP

Terraform

Kubernetes

Python

SQL

Scala

LLM/RAG Pipelines

Claude Code

AI Agents

Prompt Eng.

LangChain

GitHub Copilot

MLflow

Diff. Privacy

Unity Catalog

Data Mesh

EDUCATION

- Data Engineering Cert.**  
UC Santa Cruz, 2013–14
- B.S. Electrical Engineering**  
Sardar Patel College of Eng.
- Diploma, Electronics & Comms**  
Technical Board, Tamil Nadu

CERTIFICATIONS

- VMware SaaS Essentials
- Hadoop Fundamentals
- NoSQL Databases
- NoSQL for SQL Professionals

PUBLICATIONS

- Cloud Computing
- All About Big Data
- Future Trends in BI

AI & THOUGHT LEADERSHIP

- 160+ articles on paddyspeaks.com
- Claude Code skill development
- Vibe coding methodology
- AI agent architecture writing

OPEN-SOURCE & COMMUNITY

- Interview Studio — free Data & AI Engineering prep
- 22 data-model deep-dives + 1,500+ questions
- In-browser SQL / Python (WASM), no backend

Paddy Iyer

DATA ENGINEERING LEADER · STRATEGIC DATA ARCHITECT · AI & PRIVACY ENGINEERING · AI INNOVATOR

15+ years driving enterprise data transformation for global tech leaders across cloud/SaaS, fintech, gaming, and consumer platforms. Expert in cloud-native architectures, real-time analytics, privacy-first pipelines, and AI-augmented data operations. Proven builder of high-trust, high-performance data teams at petabyte scale. Published writer and thought leader on data architecture, AI agents, and the intersection of ancient wisdom with modern technology.

Modern Data Architecture

Privacy-First Design

Consent Architecture

AI Agents & LLM Pipelines

Prompt Engineering

Data Governance (C1–C5)

Petabyte-Scale Systems

Global Team Leadership

Data Storytelling

FEATURED EXPERIENCE

Data Engineer — Privacy & Consent Infrastructure Apr 2024 – Present  
Meta

Drove the data engineering execution of a company-critical migration moving 1,100+ advertising data assets from broad-access to consent-first architecture. Landed 800+ changes across four privacy domains. What I actually got paid for: finding the failures nobody's tooling could see

Finding the Failures the Tooling Couldn't See

- Proved the pre-deployment validation stack was structurally blind — per-table permission checks returned PASS for tables that didn't exist, and never modeled the query-level enforcement that actually runs in production. Diffs cleared every gate, then broke
- Reduced a class of migration failures to a mechanical, offline-checkable invariant: the source namespace was encoded in the old table name all along. Verified against the full asset inventory — 81% of migrations required a namespace change, not a name change. Made the correct transform regex-provable instead of lookup-dependent
- Built the hard-gate guard system that turned each production incident into a pre-land block. Every gate in the playbook traces to a specific postmortem
- Separated silent failure from loud failure — showed that the "obvious fix" for a resolution error swapped a paging error for a silently skipped subquery. Blocked it as strictly worse, and documented why
- Enforced scope integrity across a large migration batch: privacy annotations treated as invariants, not redundant syntax. Blocked collateral edits that removed safety controls to satisfy a linter

Force Multiplication

- Co-built an AI coding skill encoding the full remediation methodology — new engineers productive in 2 days instead of 2 weeks; per-asset cycle time down 3–5×
- Sustained 800+ landed changes solo — throughput normally requiring 3–4 engineers

Incident Leadership

- Owned response for 5 production failures including 2 SEV3s (one at \$5.27M/day). Fix-forward, zero reverts, grants coordinated across 6+ owning teams. Director-level escalations closed with written RCA
- Triage rule I enforced: reason from the runtime error, never from a plausible mechanism. When your analysis contradicts the pipeline owner's, you're missing a layer

**Lead Data Engineer****Meta**

Sep 2015 – Sep 2017

80+ Dataswarm pipelines for petabyte-scale ads; Cross Device Insights, Outcomes Datamart + Norms DB

**Data Engineer****GREE International**

2014 – 2015

PII masking, Vertica → Redshift, 1,000+ table optimization

**Data Architect****Chegg Inc.**

Oct 2013 – Apr 2014

**Technical Director****Model N**

2011 – 2013

Life Sciences BI, cloud migration, ETL modularization

**Architect****CallidusCloud**

2005 – 2011

Incentive comp analytics, BusinessObjects XI

**DW Architect****Hewlett Packard**

2001 – 2005

Enterprise DW + 8 datamarts; ETL 18 → 3 hrs; 40–50% sales lift via clickstream

**Data Architect / Sr. Engineer****Xoriant · Dept of Electronics, India**

1990 – 2001

**Lead Data Architect — Partner Data Engineering****VMware**

*Strategic architect for VMware's Partner Data Platform — unifying fragmented partner data into a single governed analytics layer*

**Data Architecture & Platform Unification**

- Designed and built the Partner Data Platform from the ground up, unifying fragmented partner data from 10+ disparate sources into a single governed analytics layer — eliminating data silos that had blocked cross-functional decision-making for years
- Designed an intuitive data catalog and governance model (Python/Confluence) that drove broad self-service adoption, cutting reliance on tribal knowledge by 60% and reducing ad-hoc data request volume to the data team by ~40%
- Implemented VMware's first structured data governance for partner data — classification, lineage tracking, and policy enforcement — achieving audit-ready status and enabling the team to pass compliance reviews without remediation
- Improved data consistency by 60% through standardized naming conventions, automated quality checks, and cross-team data contracts — directly reducing downstream reporting errors that had caused quarterly restatements

**AI & Performance Engineering**

- Pioneered an AI copilot for retrieval optimization, coding, and governance best practices — one of VMware's earliest LLM-assisted data engineering deployments, reducing routine query development time by ~30%
- Achieved 40%+ Spark processing time improvements through broadcast joins, caching strategies, and skew management — enabling SLA-critical reports to complete within their delivery windows for the first time
- Established automated monitoring and proactive alerting that replaced reactive firefighting, reducing pipeline incident response time from hours to minutes and cutting weekend on-call escalations significantly

**Leadership & Cross-Functional Alignment**

- Drove cross-team alignment across 5+ data teams via shared roadmaps, reducing duplicate pipeline development efforts by ~25% and accelerating delivery cycles
- Led workshops and knowledge-sharing sessions that upskilled 30+ partner data consumers, building organizational data literacy and reducing the "data team as bottleneck" pattern
- Engaged director/VP-level stakeholders to define a unified data vision, securing budget and headcount for the platform roadmap — resulting in the team growing from 3 to 8 engineers during the engagement

**Senior Consultant — Ads, Commerce & Privacy**

Jul 2019 – Dec 2021

**Meta**

*Led high-impact projects across Meta's Ads, Commerce, and Privacy teams, architecting secure, scalable, and compliant data platforms*

**Commerce Data Architecture**

- Architected a centralized data warehouse that unified product and seller data into a single analytics layer — enabling commerce teams to reduce decision-making cycle time from weeks to days and directly informing category investment strategies
- Designed and launched the Category Management Data Warehouse, providing deep cross-vertical analysis that empowered commerce teams to identify underperforming seller segments and reallocate resources, contributing to measurable GMV growth
- Owned the SMB funnel redesign, improving query performance by 3x while maintaining key functionality — enabling the pipeline to scale from thousands to tens of thousands of small business advertisers without infrastructure changes

**Privacy Engineering & Compliance**

- Led privacy remediation across 1,200+ SMB 2.0, Customer Journey, and BPO tables — resolving critical design flaws that had left the team exposed to compliance risk, and establishing remediation patterns later adopted at scale in the Safe Ads program
- Collaborated with Hive Anonymization engineers to enhance anonymization processes, enabling the ads and commerce data products to pass privacy audits without manual intervention — reducing audit preparation effort by ~60%
- Drove Salesforce ID deprecation and Facebook ID integration across 1,000+ pipelines, eliminating an external vendor dependency that had caused data consistency issues affecting ~15% of cross-system joins

**Data Platform Migration**

- Spearheaded migration of 1,000+ Hive pipelines to Spark, automating ~70% of conversion tasks — reducing what was projected as a 12-month migration to under 6 months and enabling commerce/payments teams to benefit from Spark's performance gains ahead of schedule
- Built chargeback representment and leakage dashboards (FGF, Dataswarm, Unidash) that identified ~\$XM+ in previously undetected revenue leakage, enabling the finance team to recover funds and close monitoring gaps
- Designed Marketplace App reliability dashboard covering payin/payouts and user stability — reducing MTTR for payment-related incidents by providing real-time visibility that replaced manual log analysis

## **Data Engineer Consultant**

*Dec 2017 – Apr 2018*

**LinkedIn**

- Led GDPR compliance efforts: encrypted sensitive customer data into LinkedIn Dali storage using Hive, Python, and Pig; retired legacy data sources and feeds; migrated to a stable, compliant framework

## **Lead Data Engineer Consultant**

*Sep 2015 – Sep 2017*

**Meta**

- Developed 80+ complex Dataswarm pipelines for petabyte-scale ads data across Sales Operations and Marketing Intelligence: Cross Device Insights, Global Account Pipeline, Gaming Apps Reporting, Outcomes Datamart + Norms DB, Facebook Media & Live Monetization
- Deep expertise in petabyte-scale ads data, partnering with cross-functional teams to optimize revenue generation strategies and deliver scalable analytics